Momentum

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Calculate and compare signed momentum.

SUCCESS CRITERIA - I can...

1. Multiply mass by velocity with correct units.
2. Include direction or a sign.
3. Distinguish zero total momentum from no motion.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “momentum” and “velocity” to explain your answer. Sentence starters are optional.

momentum: mass times velocity, including direction

velocity: rate of position change, with direction

mass: a measure of how strongly an object resists acceleration

01 Warm-up: bring back an earlier idea

An asteroid’s crossing window is days **13**-**17**; Earth’s is **15**-**19**. **Find the overlap** and **explain why this** is not proof of a collision.

Sentence starter: For this answer, I treated … as true. I found … . This supports … because … .

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $p = mv$; $p_{total} = p_1 + p_2$.
Use signed velocities; momentum units are kg·m/s.

Worked reasoning: A **3 kg** cart moving left at **2 m/s** has momentum **-6 kg·m/s** when right is positive. The minus sign gives direction, not a negative mass.

Your turn: Draw separate velocity and momentum arrows, labeling their different units.

Sentence starter: Both … . They differ in … because … .

Before	After

Label mass, velocity and momentum.

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

The two carts have the same speed in opposite directions. Which momentum values are positive if right is positive?

Sentence starter: I notice … . This supports … because … .



2 kg; 3 m/s right

2 kg; 3 m/s left

Marzano check - circle: ① ② ③

04 Apply the physics

A **0.50 kg** cart moves at **+6 m/s**; a **1.5 kg** cart moves at **-2 m/s**.
1. **Find each momentum** and total momentum.
2. **Draw the momentum arrows** and **explain why zero total** does not mean both carts are at rest.

Word help: at rest = not moving in the chosen reference frame

Sentence starter: For this answer, I treated … as true. I found … . This supports … because … .

Object / trial	Mass	Velocity	Momentum
A	____	____	____
B	____	____	____

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

One cart doubles mass at the same velocity. Another doubles speed at the same mass. **Compare the momentum multipliers**.

Sentence starter: Both … . They differ in … because … .

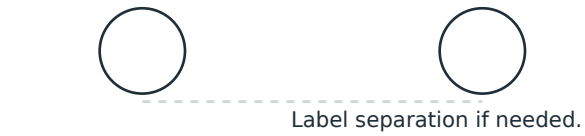
Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

Outside a spherical source, field at r is **144 N/kg**. **Find field** at **6** r and the force on **0.8 kg** there.
Label the masses, field or force relevant to the question.

Word help: field = a description of an influence at each location

Sentence starter: I found … [unit]. This value means … in this situation.



Marzano check - circle: 1 2 3

07 Exit check

A cart has momentum **-12 kg·m/s** and mass **3 kg**. **Find velocity**. Then **calculate its momentum** if velocity reverses without changing speed.

Sentence starter: I found … [unit]. This value means … in this situation.

Object / trial	Mass	Velocity	Momentum
A	____	____	____
B	____	____	____

Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

Three carts have masses **1**, **2**, **3 kg** and velocities **+4**,**-5**, v m/s. **Choose v** for zero total momentum. Then reverse only the **1 kg** cart and recalculate the total.

Sentence starter: I found … [unit]. This value means … in this situation.

Object / trial	Mass	Velocity	Momentum
A	____	____	____
B	____	____	____

Marzano check - circle: 1 2 3